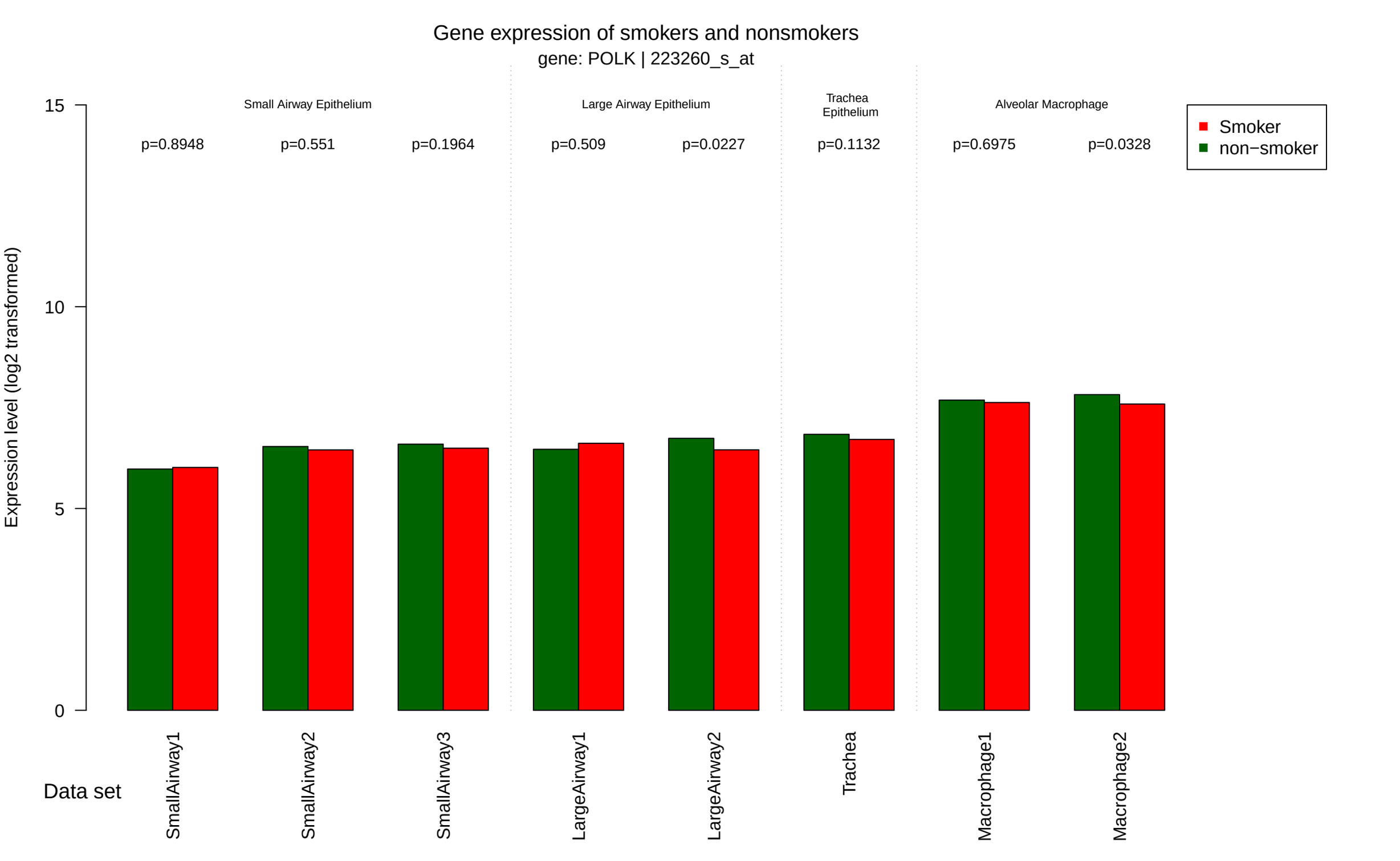
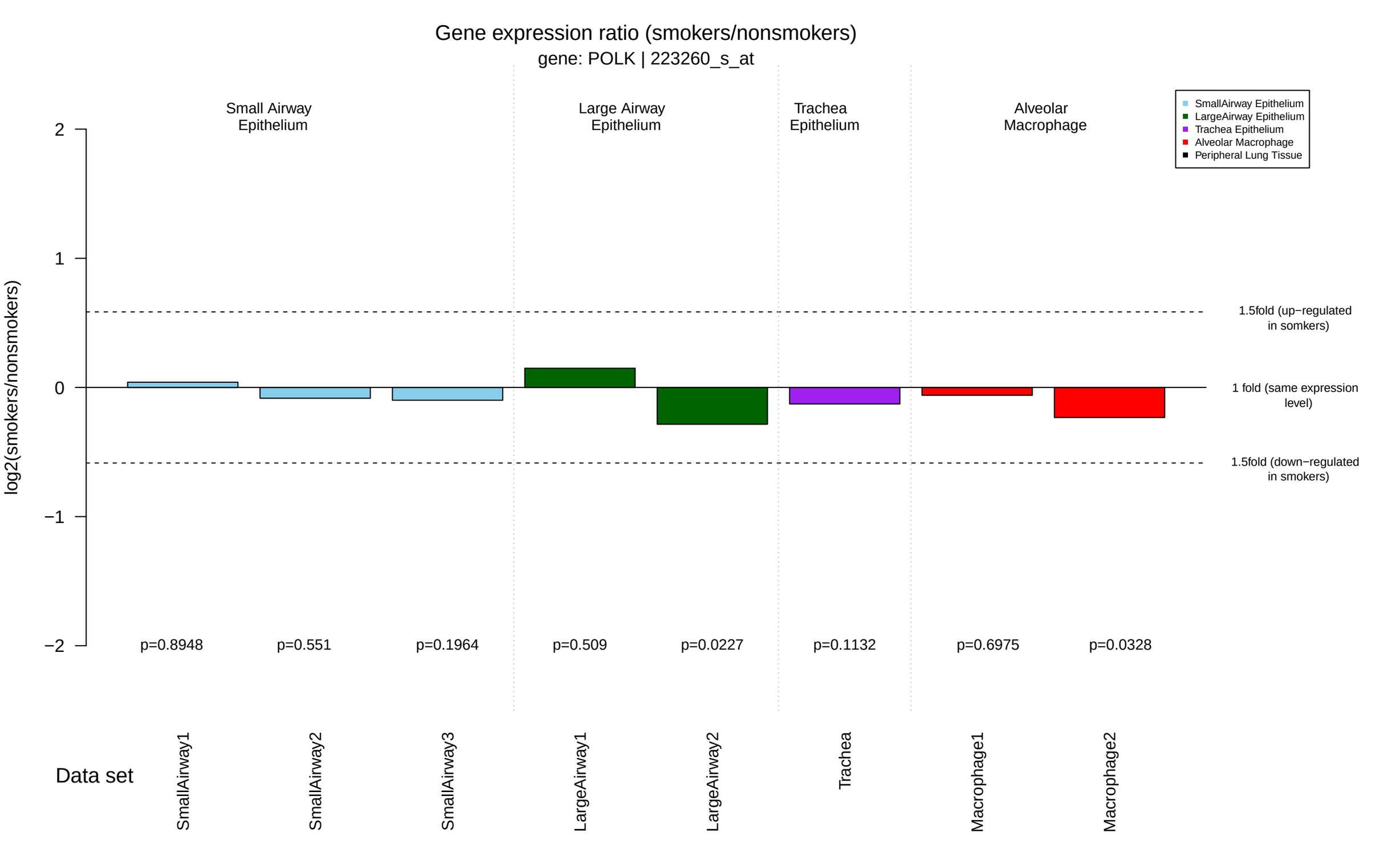
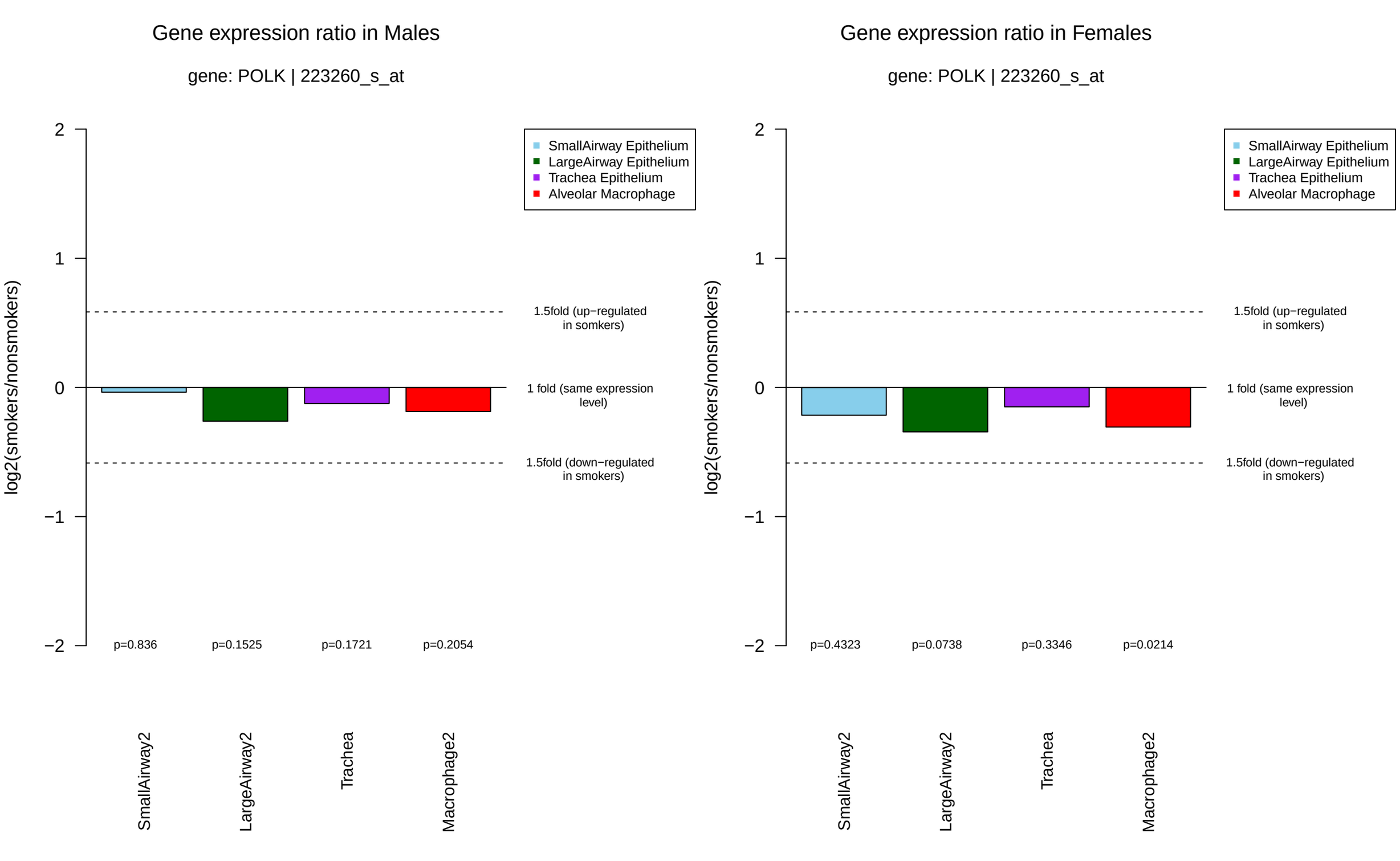
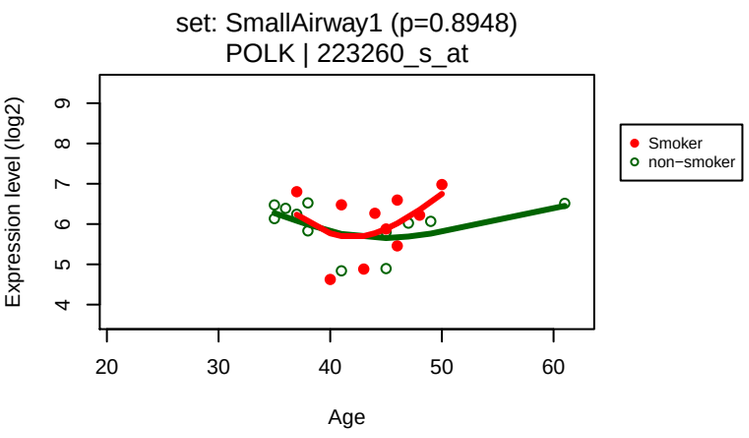
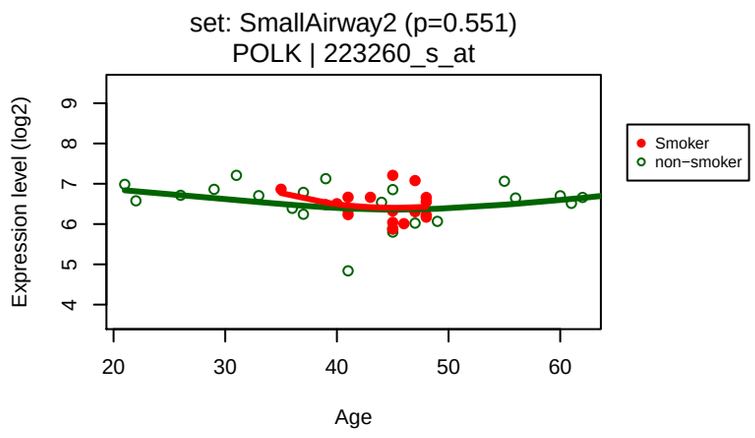
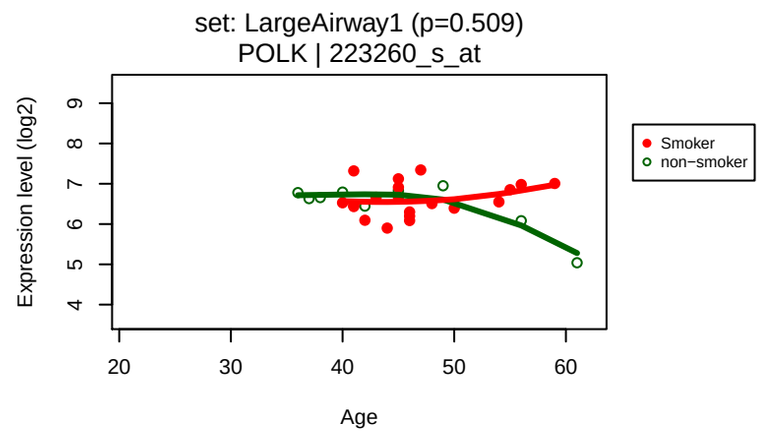

Gene expression of smokers and nonsmokers
gene: POLK | 223260_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8948
p=0.551
p=0.1964
p=0.509
p=0.0227
p=0.1132
p=0.6975
p=0.0328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

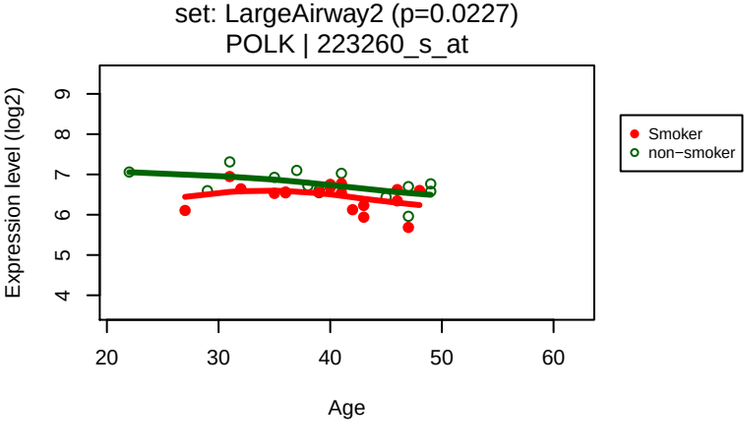

Gene expression ratio (smokers/nonsmokers)
gene: POLK | 223260_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8948
p=0.551
p=0.1964
p=0.509
p=0.0227
p=0.1132
p=0.6975
p=0.0328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

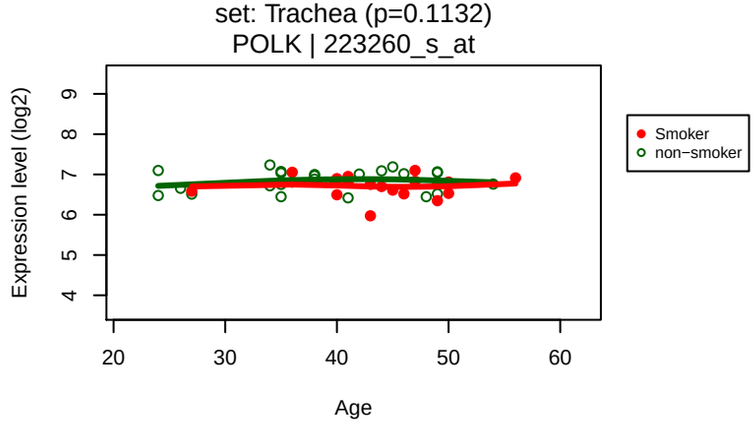

Gene expression ratio in Males
gene: POLK | 223260_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.836
p=0.1525
p=0.1721
p=0.2054
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLK | 223260_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4323
p=0.0738
p=0.3346
p=0.0214
SmallAirway2
LargeAirway2
Trachea
Macrophage2

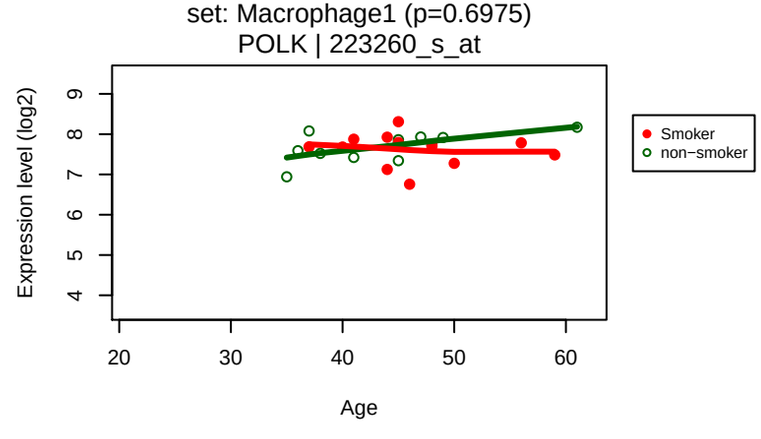

set: SmallAirway1 (p=0.8948)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker

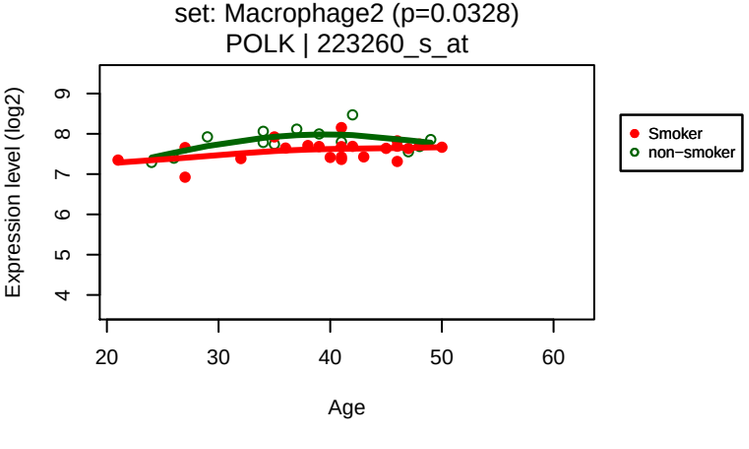

set: SmallAirway2 (p=0.551)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker

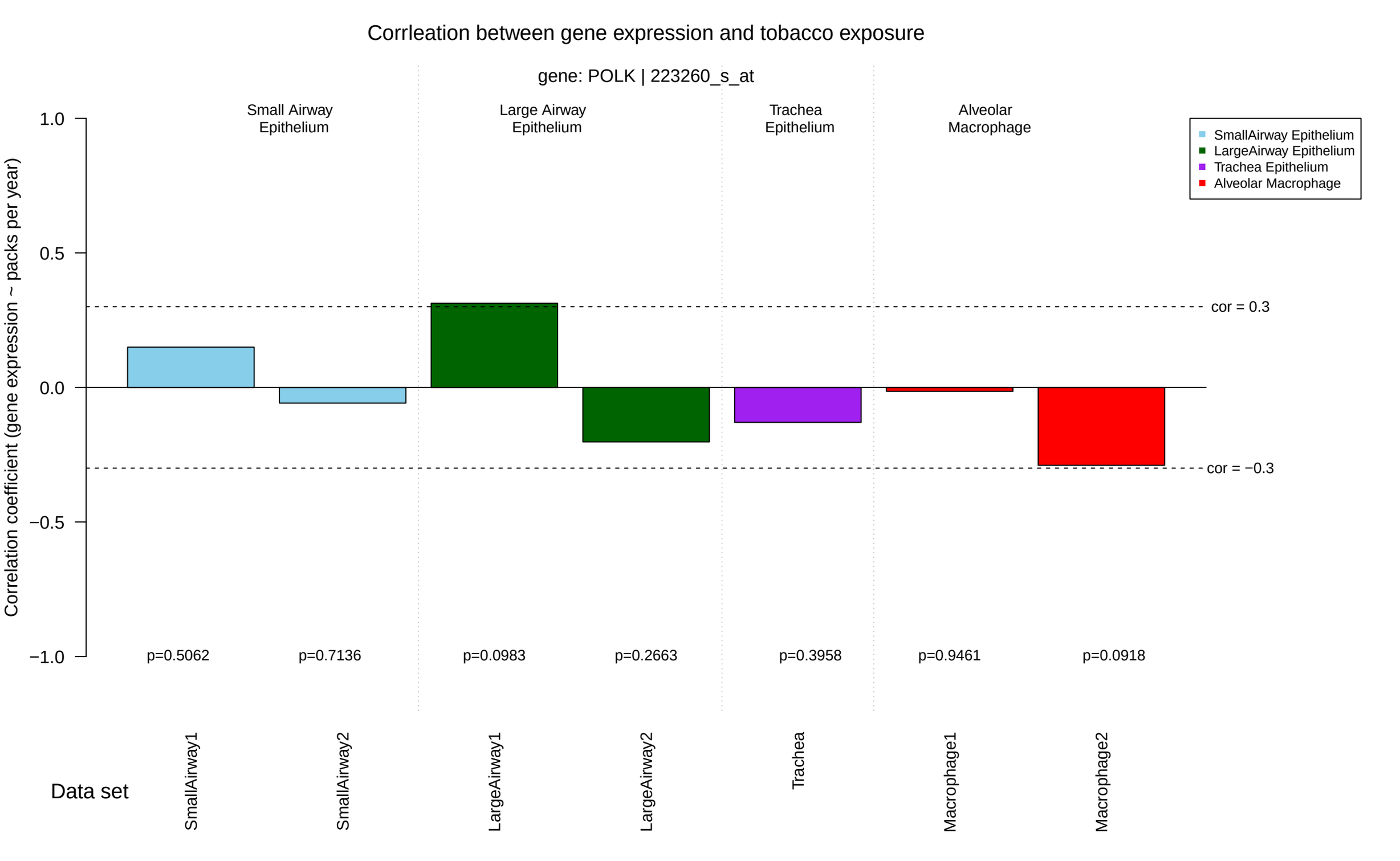

set: LargeAirway1 (p=0.509)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0227)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1132)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6975)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0328)
POLK | 223260_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLK | 223260_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5062
p=0.7136
p=0.0983
p=0.2663
p=0.3958
p=0.9461
p=0.0918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2